DO NOW — RECALL

Lesson 138 · SQL SELECT, Filtering, Sorting and JOIN

Name: _____ Class: _____ Date: _____

1. What table is created to remove an EmployeeID→DeptID→DeptName transitive dependency?

2. When does a linear search stop?

3. Which is removed first from a FIFO queue?

4. What does ROM stand for?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Department(DeptID,DeptName), referenced by Employee.

2. Model answer

On a match or after examining all items.

3. Model answer

The longest-waiting item at the front.

4. Model answer

Read-only memory.

By the end, you can…

• SELECT chooses columns from a named relation.

• WHERE restricts rows; ORDER BY arranges results.

• Link keys using INNER JOIN ... ON.

Vocabulary — know these terms

SELECT — Returns specified columns/expressions.

WHERE — Filters rows according to a condition.

ORDER BY — Sorts returned rows by chosen column/expression.

JOIN — Combines rows from related tables using a join condition.

NEW LEARNING

Read, annotate and explain

Choose projection

SELECT chooses columns from a named relation.

Filter and order

WHERE restricts rows; ORDER BY arranges results.

Join related rows

Link keys using INNER JOIN ... ON.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Orders(OrderID,CustomerID,Total) has totals 25,75,120. Write SQL listing orders over 50 with largest first.

Filter Total > 50

Find the given data and name the required result.

Select OrderID, Total

Explain why this intermediate stage is necessary.

Sort DESC 120,75

SELECT OrderID, Total FROM Orders WHERE Total > 50 ORDER BY Total DESC; returns orders with 120 then 75, excluding 25.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Write SQL listing Title and Price from Book where Price<10, sorted ascending by Price. [6]

Question 2

Explain the role of the ON clause in an INNER JOIN between Borrower and Loan. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Write an INNER JOIN query listing Student.Name and Course.Title using Student, Enrolment and Course. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

ORDER BY always removes duplicate result rows.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.